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CS 179 - Algorithms for Probabilistic and Deterministic Graphical Models
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Rediscovering the Big Five with Probabilistic Graphical Models

1. Introduction

The Big Five model of personality holds that questionnaire responses are driven by five broad traits: Extraversion, Neuroticism, Agreeableness, Conscientiousness, and Openness. We ask whether this organization into five traits emerges from raw item responses as conditional dependence structure when the model is never told that traits exist.

We use the Big Five dataset from the Open Psychometrics project (openpsychometrics.org): 19,719 respondents who rated the 50 statements of the IPIP Big Five inventory on a scale from 1 to 5. After removing incomplete records, 19,718 respondents remain, split 80/20 into 15,774 for training and 3,944 for testing. Each trait has ten items (E1-E10 through O1-O10); we use this grouping only for evaluation and never during learning.

Our primary model is a pairwise binary Markov random field (an Ising model) over the 50 items. Its structure is estimated by pseudolikelihood neighborhood selection: one logistic regression with L1 regularization per item, predicting that item from the other 49. Responses are binarized so that an answer above the neutral midpoint counts as agreement, after negatively keyed items are flipped. As a latent variable comparison we fit Gaussian factor analysis to the standardized responses. A graphical model is the right tool here because marginal correlation mixes direct and indirect relationships, while an edge in a Markov random field captures dependence that survives conditioning on every other item.

2. Resources

The dataset is BIG5.zip from openpsychometrics.org/_rawdata, which includes a codebook describing the items. The implementation is in Python and uses numpy and pandas for data handling, sklearn for logistic regression and factor analysis, networkx for community detection, and matplotlib for figures. The structure learning approach follows Ravikumar, Wainwright, and Lafferty (2010). Factor loadings are shown after varimax rotation, which changes only their presentation and not the fitted model. All pipeline code (preprocessing, the two models, evaluation, and the experiment runner) was written for this project.

3. Evaluation and Findings

We evaluate on the 3,944 test respondents with two predictive metrics and two structural ones. The first predictive metric is the conditional log likelihood: the mean, over test respondents and items, of the log probability of each answer given the other 49. This is the natural test analogue of the pseudolikelihood objective used for training. The second is masked item accuracy: the fraction of answers predicted correctly from the other 49. The structural metrics ask how much of the learned graph respects the trait blocks: the share of total edge weight that falls inside trait blocks, where random pairing would give 18.4 percent, and the agreement between graph communities (greedy modularity on absolute edge weights) and the designed trait labels, measured by the adjusted Rand index (ARI). An independent baseline that uses only item frequencies serves as the predictive reference.

3.1 The learned graph recovers the Big Five

At the default regularization (C = 0.1) the model selects 1,002 edges. Figure 1 shows the absolute edge weights with items ordered by trait, and a clear block pattern appears along the diagonal. Quantitatively, 54.3 percent of all edge weight lies inside trait blocks, three times the 18.4 percent chance rate. The edge count tells a more nuanced story: only 21.7 percent of edges connect items of the same trait, barely above chance. Together these numbers say that the graph contains many weak edges between traits but concentrates its strong dependencies inside traits. The weight, not the count, carries the structure.

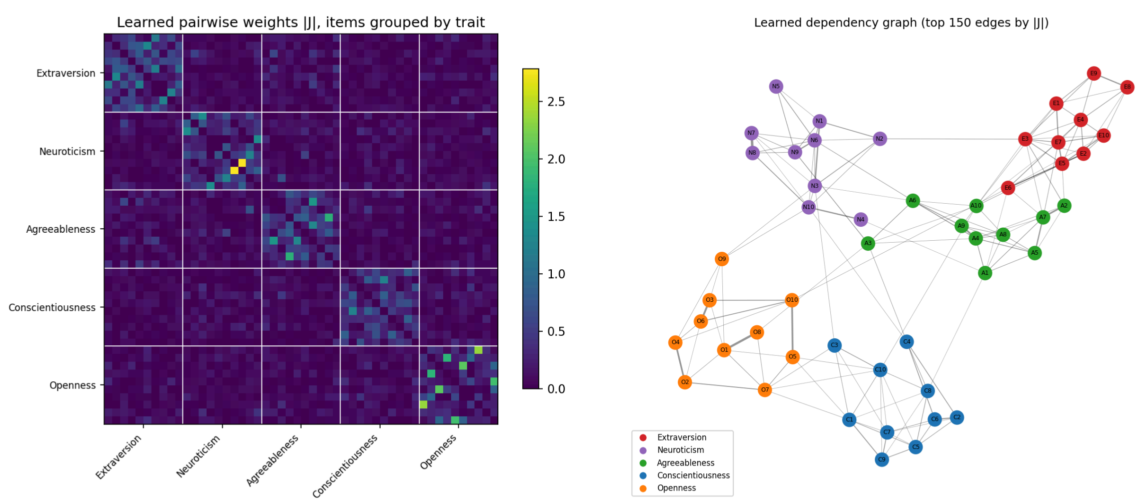


Figure 1: Absolute learned pairwise weights with items ordered by trait.

The structural headline is community detection. Greedy modularity on the learned graph yields exactly five communities, and every one of the 50 items lands in the community of its designed trait, giving ARI = 1.00. The organization of the inventory into five traits is recovered perfectly from raw binary responses by a model that was never told traits exist (Figure 2).

Figure 2: The learned dependency graph with nodes colored by trait.

3.2 Learned dependencies improve prediction

The Ising model reaches a test conditional log likelihood of −0.471 against −0.635 for the independent baseline, and a masked item accuracy of 77.6 percent against 64.5 percent.

Conditioning on a respondent's other answers substantially improves prediction of an unseen answer, which confirms that the learned edges capture real dependence rather than noise.

3.3 Effect of training set size

Figure 3 shows the learning curve. With only 250 training respondents the model already reaches 73.5 percent accuracy; 2,500 respondents give 77.2 percent, and the full 15,774 give 77.6 percent. The conditional log likelihood behaves the same way: −0.554 at 250 respondents, −0.473 by 5,000, and −0.471 at the full size. The dependence structure of 50 items is largely learnable from one to two thousand respondents, with strongly diminishing returns beyond that point.

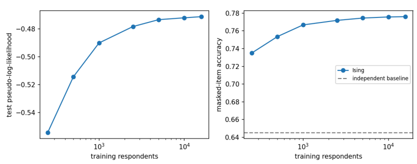


Figure 3: Test conditional log likelihood and masked item accuracy as the training set grows.

3.4 Regularization and the cost of sparsity

Varying the inverse regularization strength C from 0.001 to 1.0 grows the graph from 29 edges to 1,192 (Figure 4). Predictive performance saturates early: accuracy reaches 77.6 percent and the conditional log likelihood reaches −0.4713 by C = 0.1, and C = 0.316 is essentially identical. Interpretability moves in the opposite direction: the share of edge weight inside trait blocks falls from 93.7 percent at the sparsest setting to 49.9 percent at the densest. In other words, the strongest edges are almost exclusively within traits, and the weak edges added under light regularization mostly connect different traits while barely helping prediction. An aggressively sparse model is nearly pure trait structure at a modest cost in accuracy: 73.3 percent with 29 edges.

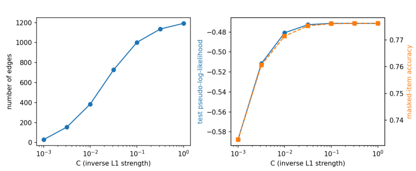
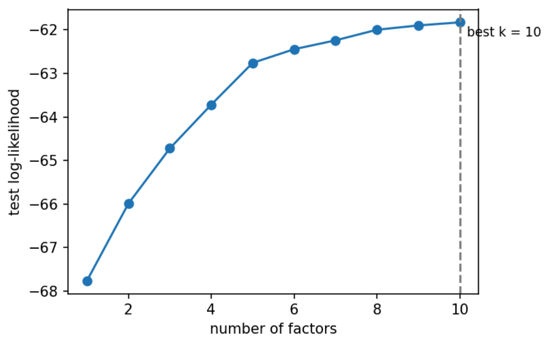


Figure 4: Edge count and test performance across the regularization path.

3.5 How many latent factors does the data support?

For the factor analysis comparison we sweep the number of factors k from 1 to 10 and measure the test log likelihood (Figure 5). The likelihood improves monotonically through k = 10, so there is no peak in the tested range, and by raw maximum the best k is 10. The marginal gains, however, collapse after five factors: the fifth factor improves the test log likelihood by 0.96, the sixth by only 0.31, and each later factor by 0.24 or less. The data therefore supports the Big Five as the dominant shared structure, with later factors contributing small refinements, plausibly nuances among items inside a trait, rather than evidence for a competing trait count. After varimax rotation, each of the five trait blocks in the loadings is dominated by a single distinct factor (Figure 6). The order and signs of factors are arbitrary in factor analysis; what matters is that blocks and factors align one to one.



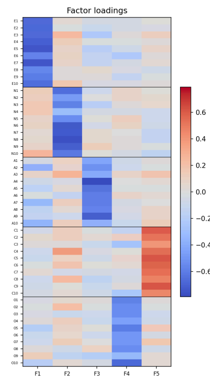


Figure 5: Test log likelihood as a function of the number of factors.
Figure 6: Factor loadings at k = 5 after varimax rotation.

3.6 Robustness to the binarization rule

Because converting responses on a 1-5 scale into binary values is a modeling choice, we repeat the main analysis with a threshold at each item's training median in place of the agreement rule. The results are essentially unchanged: 1,021 edges against 1,002, accuracy of 77.2 percent against 77.6 percent, a within block weight share of 52.9 percent against 54.3 percent, and community recovery remains perfect with ARI = 1.00 under both rules. The conclusions do not hinge on the threshold.

3.7 Direct dependence against latent causes

Finally we compare the two model classes on the same task: predicting each masked answer in the binary matrix. The Ising model reaches 77.6 percent accuracy, factor analysis reaches 76.8 percent, and the independent baseline reaches 64.5 percent. Two very different explanations of the data, direct pairwise dependence and a small set of latent causes, converge on nearly identical predictive performance and on the same account of five groups.

4. Conclusion

Starting from raw survey responses, an Ising model with L1 structure learning rediscovers the Big Five: graph communities match the designed traits perfectly (ARI = 1.00), the share of edge weight inside trait blocks is three times chance, and the learned dependencies improve test accuracy by 13 points over an independent baseline. The regularization path reveals a clean tradeoff: sparse graphs are almost pure trait structure, while dense graphs add weak edges between traits with negligible predictive value. The factor analysis comparison independently points to five dominant factors through a sharp elbow in the test likelihood and through rotated loadings that assign each trait block its own factor. Limitations: binarization discards ordinal information, Gaussian factor analysis is an approximation for ordinal data, and the items were designed to cluster into these traits, so our result confirms the structure of the instrument rather than discovering personality from neutral text. A natural extension is an ordinal or Gaussian copula graphical model fit directly to the original responses.